

Understanding Customer Experience & Retention Drivers in Brazilian E-commerce

A Data-Driven Analysis Using the Olist Brazilian E-commerce Dataset

Prepared By:

Deepak Pathak

Role:

Data Analyst Case Study

Dataset:

Olist Brazilian E-commerce Public Dataset

Tools Used:

Python, Pandas, Plotly, Scikit learn

Introduction:

This analysis explores customer behaviors, delivery performance, spending patterns, and customer satisfaction using the Olist Brazilian E-commerce dataset. The objective is to identify operational and customer-related factors influencing retention, revenue generation, and overall customer experience.

KPI Section:

Total Order: 99441

Total Customers: 96096

Total Revenue: 20579664.01

Average Order Value: 172.74

Average Review Score: 4.02

Average Delivery Days: 12.02

Customer Behaviour & EDA:

Section 1: Monthly Revenue Trend

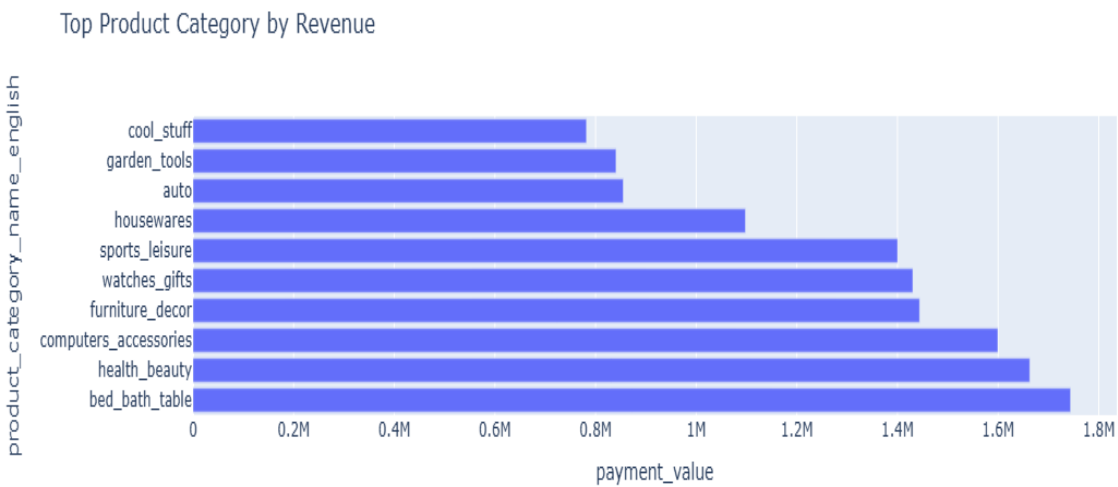
Monthly Revenue Chart



Insights:

Revenue demonstrates seasonal growth patterns with noticeable spikes during peak shopping periods.

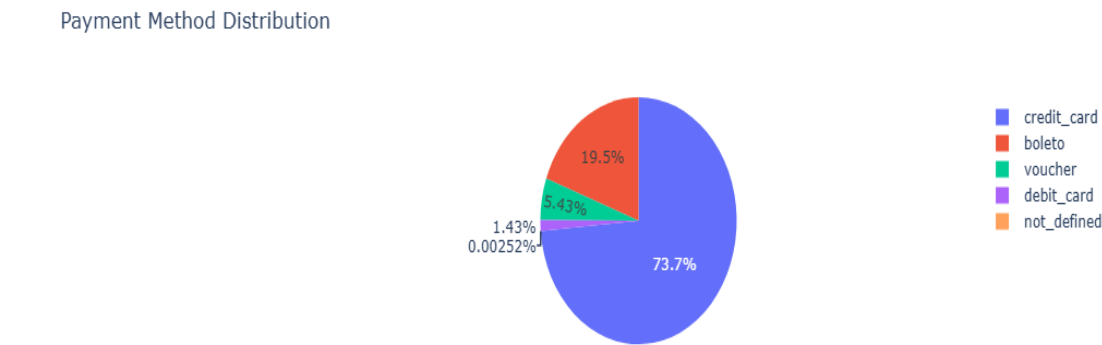
Section 2: Top Product Categories



Insights:

A small number of product categories contribute a disproportionately large share of overall revenue.

Section 3: Payment Method Analysis

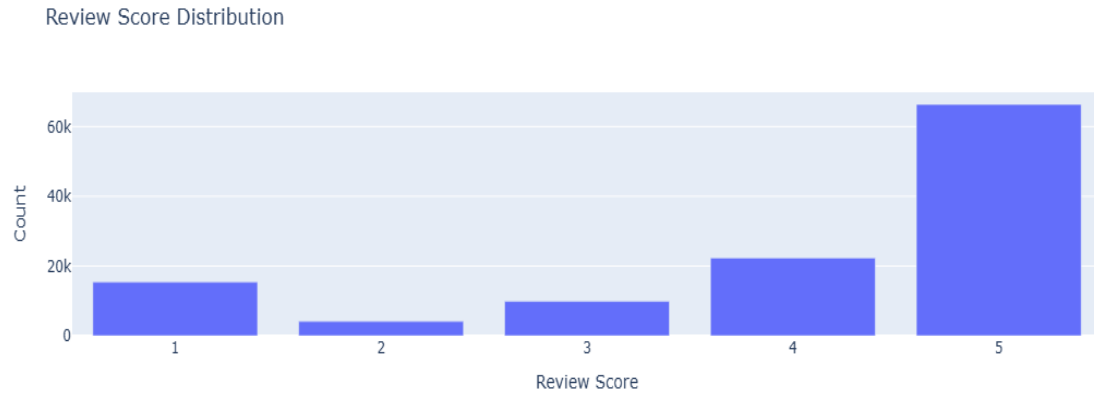


Insights:

Credit cards dominate customer transactions, indicating strong preference for installment-based payment methods.

Delivery Experience & Customer Satisfaction:

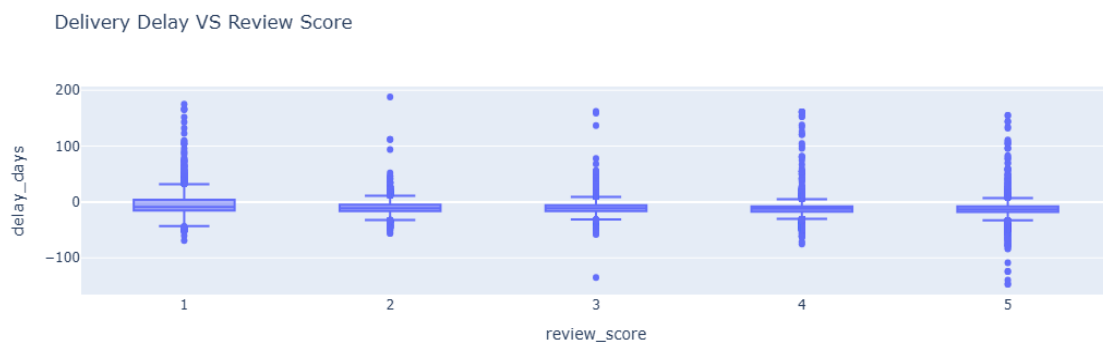
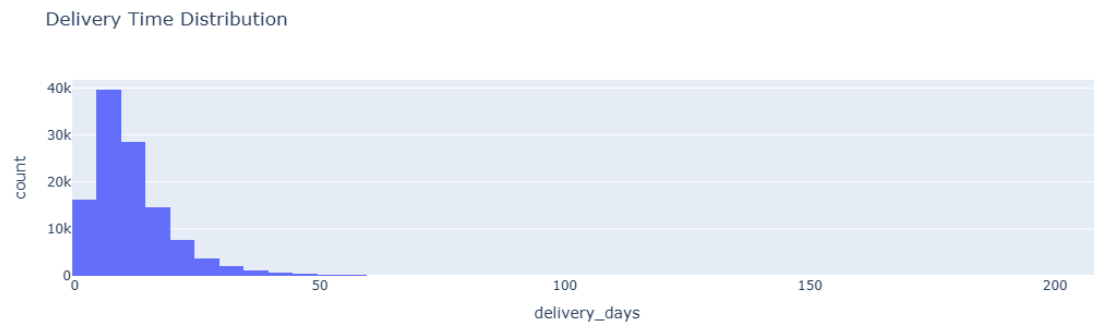
Section 1 : Review Score Distribution



Insights:

Customer reviews are highly polarized, with a large concentration of both 5-star and 1-star ratings.

Section 2: Delivery Delay Analysis



Insights:

1. Most deliveries are completed within a moderate timeframe, though some extreme delays exist.
2. Lower review scores are strongly associated with higher delivery delays.

Hypothesis Testing & Recommendations

Hypothesis 1:

Late Deliveries Reduce Customer Satisfaction

Objective

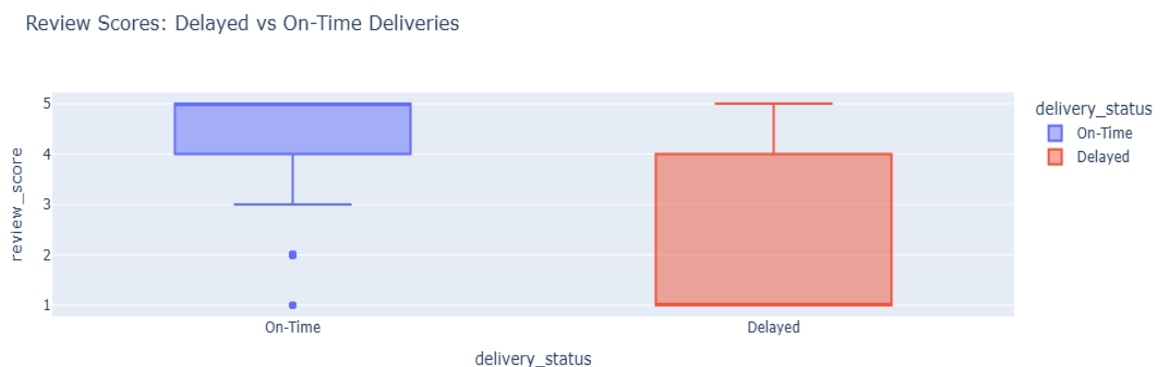
The objective of this analysis is to examine whether delayed deliveries negatively impact customer satisfaction levels on the e-commerce platform.

Customer satisfaction is measured using review scores provided after order delivery.

Statistical Test Used

An Independent Two-Sample T-Test was performed to compare review scores between delayed and on-time deliveries.

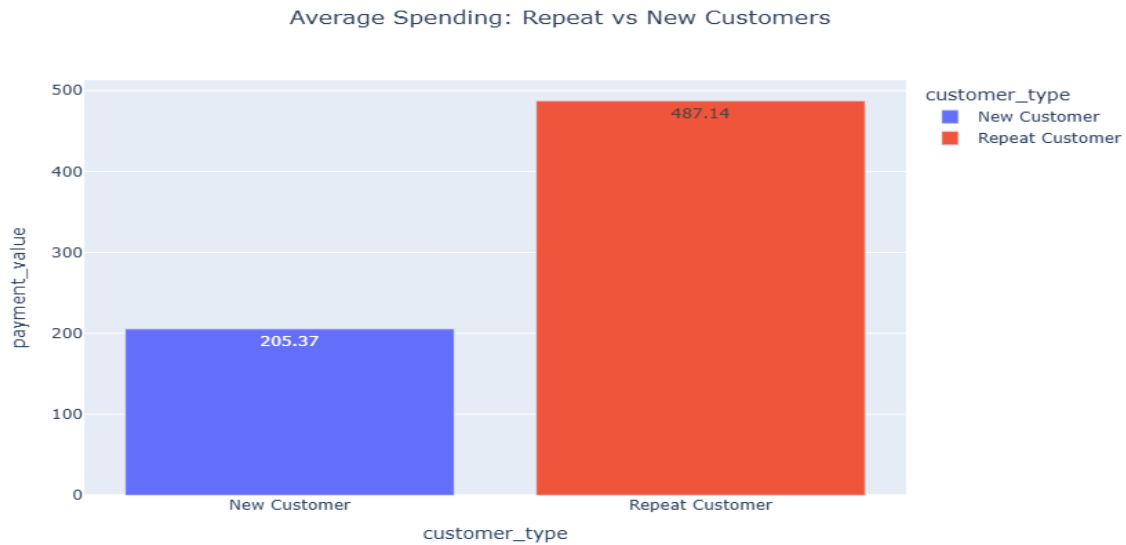
p-value = 0.0 for this test



Insights:

1. Orders delivered late received significantly lower review scores compared to on-time deliveries.
2. Customers are highly sensitive to delivery experience, making logistics performance a critical factor in customer satisfaction.





Insights:

Repeat customers spend significantly more compared to first-time customers.

Although repeat customers represent a smaller proportion of total customers, they contribute substantially higher revenue per customer.

The analysis highlights the strong long-term value of retained customers.

Repeat customer spending is more than twice that of new customers, demonstrating the importance of retention-focused business strategies.

Actionables

Actionable 1: Improve Delivery Performance in High-Delay Areas

Evidence

The analysis revealed that delayed deliveries significantly reduce customer satisfaction levels. Customers receiving delayed orders consistently provided lower review scores compared to customers receiving on-time deliveries.

The statistical test also confirmed a significant relationship between delivery delays and customer satisfaction ($p < 0.05$).

Recommendation

The company should prioritize logistics optimization in regions and seller networks experiencing frequent delivery delays.

Possible actions include:

- Improving warehouse allocation
- Optimizing shipping routes
- Partnering with faster logistics providers
- Monitoring high-delay sellers more closely

Expected Business Impact

Reducing delivery delays can improve customer satisfaction, minimize negative reviews, and increase the likelihood of repeat purchases.

Actionable 2: Strengthen Customer Retention Strategies

Evidence

Repeat customers demonstrated significantly higher spending behavior compared to first-time customers.

Average spending among repeat customers was more than twice that of new customers:

- New Customers: 205.37
- Repeat Customers: 487.14

The statistical analysis also confirmed this difference to be significant ($p < 0.05$).

Recommendation

The platform should invest in retention-focused initiatives such as:

- Loyalty and rewards programs
- Personalized recommendations
- Email remarketing campaigns
- Exclusive offers for returning customers

Expected Business Impact

Increasing customer retention can significantly improve customer lifetime value (CLV) and long-term revenue growth without depending entirely on new customer acquisition.

Further Exploration:

1. Customer Churn Prediction
2. Delivery Delay Prediction

Appendix

LLM Usage Disclosure

Tools Used

ChatGPT (OpenAI)

Areas Where LLM Assistance Was Used

LLM assistance was used for:

1. Python syntax support
2. Plotly visualization guidance

Areas Independently Completed

The following components were independently analyzed and curated:

1. Business insights
2. Hypothesis formulation
3. Statistical interpretation
4. Recommendations and actionables
5. Final business storytelling and report structure

Conclusion

1. This analysis explored customer behavior, delivery performance, spending patterns, and customer satisfaction within the Olist e-commerce platform.
2. The findings revealed that delivery delays significantly reduce customer satisfaction, while repeat customers contribute substantially higher revenue compared to first-time customers.
3. The study highlights that improving logistics efficiency and strengthening customer retention strategies can create significant long-term business value for the platform.